CSC 123 Programming and Problem Solving II

California State University Dominguez Hills
Department of Computer Science and Technology

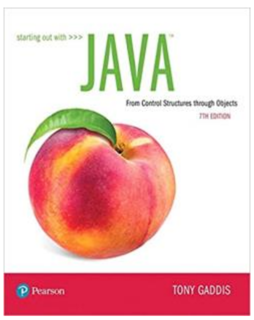
Course:	**CSC 123 Programming and Problem Solving II**			
Sections:	07 (lecture)			
	08 (lab)			
Time:	**T/Th**	Lecture	10:00-11:15	II-3310
		lab	11:30 - 12:45	II-3310

Professor: **Malcolm McCullough**
Email: **mmccullough@csudh.edu**
Office Hours: **T** **1:00-3:30** **NSM B-104** **and by appointment**

Description:
*In this course we will discuss the principal concepts of object-oriented programming in Java. We will provide a thorough conceptual grounding in object-oriented programming techniques and strategies, program design with objects and classes**, inheritance and polymorphism, encapsulation and information hiding, exception handling, text processing, file processing, recursion, the object-oriented programming as a paradigm and** graphical interface design* and *event-driven programming (as time permits**,**.*

Prerequisites:
- CSC 121 with grade "C" or better
- Consent of Instructor.

Required Materials
- **Textbooks:**



[Required]
Starting Out with Java: From Control Structures through Objects (7th Edition)
Author: Tony Gaddis,
Publisher: Pearson (February 26, 2019)
ISBN-10: 0134802217   ISBN-13: 978-0134802213

- **Hardware - Laptop/Desktop**:
  - CPU: i5 6th generation (or equivalent); RAM: 8 GB; HDD: 1 GB of free disk space
  - Webcam: A built-in or standalone webcam. Respondus, will not allow your smartphone to be as a web cam.
  - Software**:**
    - Microsoft or Unix OS with typical office and web browser tools and administrative or *root* privileges
    - Java SDK - SE 14 or newer
  - see ***Minimum Computer Specifications*** below

Time commitment:
*This is a 4-unit course. Which means it has **3 hours of lecture per week, 3 hours of laboratory per week** and up to an **additional 12 hours** per week (course activities, projects, reading, studying, etc). That is**, it is expected that you can spend 18 hours each week** on just this course.   If you are not going to be able to make the necessary time with your current schedule (work, family, and school) you will very likely not successfully complete this course.*

I understand the time commitment needed for this course and agree to allocate the time needed to successfully complete this course __T.E.C.__ initials/date.

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Course Goals:
The goal of the course is to
- Enhance the students basic programming skills, especially program control structures
- Build a foundation to be able to understand concepts of object-oriented principles and their implementations in Java.
- Provided the student with computational problem solving techniques based on object-oriented programming using Java, including program design with objects and classes, inheritance and polymorphisms, encapsulation and information hiding, exceptions and handling, stream I/O and file processing, interface design and implementation using AWT and Swing, event and event listeners, and recursion.

Learning Objectives:
Program Learning Outcomes (PLOs):
1. **Analyze a complex computing problem** and to apply principles of computing and other relevant disciplines to identify solutions.
2. **Design, implement, and evaluate** a computing-based solution to meet a given set of **computing requirements** in the context of the program's discipline.
3. **Communicate effectively** in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Course Outcomes:
Upon successful completion of this course a student should:
- understand and use **object-oriented programming concepts** including classes, **objects, inheritance, polymorphisms**, and information hiding;
- design Java programs with classes, methods, and objects;
- identify classes and their attributes and methods from the problem description;
- master object-oriented programming skills and strategies such as class hierarchy and class diagrams;
- understand and use exceptions and exception handling as well as file processing;
- understand wrapper classes and be able to develop text processing programs in Java;
- be able to design and implement graphical user interfaces (GUI) using JavaFX;
- master event-driven programming and develop event listeners in Java;
- understand recursion and be able to develop recursive methods.

Standards of Student Conduct
All students must conform to the Standards of Student Conduct, which have been established by students and college staff and have been approved by the Board of Trustees. The Standards of Student Conduct are listed in the Academic Policies section of the university Catalog.

Behavioral Standards
Behavior (which includes course communication and dialog) that persistently or grossly interferes with classroom activities is considered disruptive behavior and may be subject to disciplinary action. Such behavior inhibits other students' ability to learn and an instructor's ability to teach. The instructor may require a student responsible for disruptive behavior to leave class pending discussion and resolution of the problem and may also report a disruptive student to the Student Affairs Office (WH A-410, 310-243-3784) for disciplinary action.

Communication Etiquette: *In addition, when you send an email to your instructor, you should:*
- *ALWAYS use the university campus email system TORO email. Do not use Canvas email/messaging system*
  - messages sent via Canvas are supposed to be forward to campus email (but it does nto always work)
- *ALWAYS put **COURSE NUMBER** and possibly section number) in the subject line.*
- *Be brief and concise but still give context for question or issue.*
- *Avoid attachments*
- *Never send course work (assignments) via email*
- *Emails sent Monday through Thursday, typically will be respond to within 36 hours; Messages sent Friday (to Sunday) may not be read until Monday.*

Netiquette: *When communicating online, you should always:*
- *Treat others with respect. Always be respectful of others' event if you disagree (try to see from another point of view)*
- *Be cautious using humor or sarcasm; tone is usually lost, and you can easily give offensive.*
- *Use clear, concise, and professional language. (Avoid overuse of emoticons like 😀).*
- *Use standard fonts and font sizes (10/12 pt. font)*
- *Avoid using the all caps (cap-lock) - IT IS INTERPRETTED AS YELLING.*
- *Be careful with personal information.*

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Course Expectations:
*Students are expected to study the material presented in lecture. They are expected to read the chapters and sections corresponding to the lecture material, and read all assigned readings. Students are expected to complete assigned homework and projects in-depth and thoughtful answer, with alacrity. Students are expected keep up with current material, previewing course material, reviewing and rewriting course notes. Note, students should always write their own course notes (relying on presentation slides is too passive). Students are responsible for all material presented in class regardless of attendance. If you miss a lecture watch the recored version (if available) AND ask a classmate to share their notes. It is extremely important not to get behind in your work. Course material is much more easily understood by reading, practicing, and reviewing the material **over time**.*

The method of instruction shall be discussions of course materials and then, when possible, class activities. Students are encouraged to ask questions and to requests additional explanation when you are uncertain about concepts or other items discussed in this course.

Each course unit is corresponding to one hour of lecture and three hours of course work a week. Thus a 3-unit course has a time commitment of 12 hours and a 4 unit course has a time commitment of 18 hours a week (4+12=18). If you cannot make this time commitment it is recommended that you do not try to take this course.

Course Policies:
- *Students responsible for all material presented and from readings, assignments, etc regardless of for attendance or participation.*
- *Deliverables (assignments homework, quizzes, and projects) are generally **not** accepted past the due date.*
- *Deliverables (assignments homework, quizzes, and projects) submitted after the due date may receive a score of 0.*
- ***Any exceptional, non-academic, unforeseeable circumstances should be brought to the instructor attention as soon as they arise or is reasonably possible**. **Unless it is infeasible this needs to be done prior to the due date of the deliverable or exam. Failure to notify the instructor of the issue(s), as soon as reasonable possible, forfeit your right to any special accommodations.***
- *The instructor reserves the right to award partial credit for deliverables that are incomplete, haphazard, sloppy, or for project code - do not work. Partial credit is awarded solely at the instructor's discretion, and only for work that merits such an award. Assignments that are incomplete, sloppy, slipshod, or incongruous with the specifications can or may be returned to the student ungraded (score of 0).*

Academic Integrity
*Academic integrity is of central importance in this and every other course at CSUDH. You are obliged to consult the appropriate sections of the University Catalog and obey all rules and regulations imposed by the University relevant to its lawful missions, processes, and functions. In addition, you are required to read a follow the course Academic Honesty Pledge. In essence, **all work turned in by a student must be the students' own work**. Plagiarism and or other forms of academic dishonesty or cheating cannot be tolerated and will be dealt with according to course Academic Honesty Pledge. **The consequences for being caught plagiarizing, academic dishonesty, or cheating range from a minimum of zero points awarded for questionable work to receiving an F for the course grade, to recommendation for expulsion from the University.** Helping your classmates in commendable and applauded; be careful give actual working code to others is academic dishonest. Show how something works or how it can be done abstractly or with pseudo code*
Participation Policy:
Students are expected to participate fully. Participation has an indirect but significant impact on your final course grade.

Substantive participation is defined as active involvement in-class and discussion board activities (questions/responses/statements that are rich, deep and probing). It may include piggybacks on someone else's comment, challenging assumptions or adding depth to the discussion. Sometimes it is a new idea or question. Substantive input adds depth to a discussion and carries its own weight. It demonstrates that students are using his/her critical thinking skills and values the advancement of knowledge for themselves and others.

Students are strongly encouraged to ask relevant questions, make pertinent comments, and present answers to questions on the class discussion board. Course announcements will be made during class, and at a later time may be posted on Canvas's announcements. Students are encouraged to use email to communicate with the instructor on individual matters related to the course. It is the student's responsibility to ensure that the e-mail provided is correct, or that the e-mails are forward to an address that students check daily.
Knowing Your Responsibilities
CSUDH provides the student with a wide variety of academic assistance and support, but it is up to the student to know when they need help and to seek it out. It is their responsibility to keep informed and to obey the rules, regulations and policies which control their academic standing and life as a CSUDH student. Meeting deadlines, completing prerequisites and satisfying the degree and certificates requirements, as found in the curriculum guides in this catalog, are all part of the duties as a student. Consult this catalog, the college and school announcements and the schedule of classes for the information needed. Watch for official announcements.

Withdrawal from Class Policy:
The administration of this institution has set deadlines for withdrawal of any college-level courses. These dates and times are published in that semester's course catalog. Administration procedures must be followed. It is the student's responsibility to handle withdrawal requirements from any class. In other words, the instructor cannot drop or withdraw any student once enrolled in this course, after the instructor drop date as noted in the course catalog. Students must complete the appropriate paperwork to ensure that he/she will not receive a final grade of Withdrawal Unauthorized "WU", which is equivalent to an "F" in this course.

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Computer Information Literacy Expectations (*Computer Literacy Skill)*
It is expected that students will:

1. *have an understanding of basic computer hardware and software.*
2. *have access to a personal computer (see specification below) and have administrative or root privileges.*
3. *have access to Internet.*
4. ***be able to download and install software.***
5. ***be familiar with (able to use) software developing tools (IDE).***
6. *be familiar with using email as a communication tool and check campus daily (always check before star of lecture).*
7. *be able to access course websites (Canvas) and check the course site often.*
8. *be to use a word processing and other office like program.*

Minimum Computer Specifications: *In addition to the hardware and software requirements listed above, there are many software tools that will be used to deliver content, discussion, and administer exams. Students must have a computer system and Internet access that is compatible with Canvas, the campus' Learning Management System and Zoom, a video conferencing platform. To ensure reasonable interactive sessions, all students must have, at a minimum, the following specifications on their personal computers:*

System Requirements:

- *Operating System: Windows 10, MacOS 10.13+, Linux with 4.4+ Kernel*
- *CPU: i5 6th generation (mid-range performance 2015); RAM: 8 GB; HDD: 1 GB of free disk space*
- *User and administrative account (Administrative or root privileges)*
- *Audio / Video: web-camera, speakers and microphone*
- *Internet browser: Safari / Chrome / Firefox or Explorer (with current updates)*
  - *Respondus Lockdown Browser: Required for quizzes/exams.*
- *Stable Internet access (minimum 10 Mbps)*
- Java SDK (Java SE 14 or better); useful videos:

System software or platforms used

- Java SDK (Java SE 14 or better); useful videos:
  - How to install multiple java versions on MacOS
  - How to install JDK on Ubuntu
- ***Canvas's Learning Management System:*** *You can access much of the course martial through Canvas https://canvas.csudh.edu/. .*

Technical Support and Campus Resources

- *Campus Service requests*
  - *Campus Resources & Services*
- *IT support and Knowledgebase*
  - *Campus IT Support*
- *Student disAbility Resource Center – helps students with disabilities have full access to the university's educational, cultural, social, and physical facilities and programs.*
  - *Disabled Student Services*
- *Learning / Tutoring / Testing Center. Has both one-on-one and group sessions*
  - *Learning Center (Tutoring Center)*
- *Help with using Zoom video Conferencing software*
  - *Zoom Tutorials*
- ***Technology Loaner Program***
  If a student does not have access to a system with the above requirements, they should contact
  - ***CSUDH Technology Loaner Program*** *(https://techloaner.csudh.edu/)*
    Laptops, webcams, headset, and mobile Internet access (Mifi) devices

Special Needs
Online courses are required to meet ADA accessibility guidelines. This means that all aspects of the online learning experience are accessible. Please let me know if you have adaptive software and hardware to assist you with taking this course or if you have any specific needs I should be aware of. The CSUDH Student Disability Resource Center (SdRC) is available to assist you during this course. The SdRC is available at (310) 243-3660 and can be reached by email at dss@csudh.edu.

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Americans With Disabilities Act:
CSUDH adheres to all applicable federal, state, and local laws, regulations, and guidelines with respect to providing reasonable accommodations for students with temporary and permanent disabilities. If you have a disability that may adversely affect your work in this class, you are encouraged to register with Student disAbility Resource Center (SdRC) office and talk to your instructors about how to get the most out of the course and how they can help. All disclosures of disabilities will be confidential. NOTE: no accommodation can be made until you register with the SdRC.. The SdRC is committed to providing all of the University educational, cultural, social and physical facilities and programs available to students with disabilities. The program serves as a centralized source of information for students with disabilities and those who work with them. By providing support services, SdRC assists students with disabilities in the enhancement of their academic, career and personal development. The SdRC Office is located in WH D-180 phone 310-243-3660 (voice) or 310-243-2028 (TDD). Please refer to the SdRC Handbook or website https://www.csudh.edu/sdrc/ for more information.

Due Dates and Make-up Work Policy:
The completion of assignments, homework, course activities, quizzes and programming projects are mandatory and crucial to mastery of the subject matter of this class. Student studying requirements encompass previewing and reviewing, chapters, as well as lecture PowerPoint presentations; homework assignments; activities; programming projects; quizzes; and exams. The due date for homework assignments; activities; programming projects and quizzes will be posted on Canvas. No credit will be given for untimely submission of homework assignments, project, etc., except under exceptional, non-academic, unforeseeable circumstances that were discussed with the instructor as soon as they arise and prior to the due date of the deliverable or exam or as soon as reasonably possible. Failure to notify the instructor such circumstances in a timely manner forfeits any right to any special accommodations.

Make-up examinations and quizzes:
The midterm exam made be retaken at the end of the semester, at the same time as the Final exam. There is no make up for the final exam.
Make-up quizzes will NOT be given under any circumstances.

Exams: There will be TWO exams, a midterm exam and a final exam. These exams will be given during the second meeting of the 8th, week and on the final exam week (see the final exam schedule posted by university https://www.csudh.edu/class-schedule/fa24/final-exam). The exams will cover the most recent material from the book, lectures, assignments, projects, quizzes, homework, and any other activities. The final exam will NOT be compressive.

Quizzes: There will be quizzes/activities assigned for most weeks of the semester. They will be on the material most recent conveyed in lecture assigned reading or project. These quizzes activities will be asynchronous, that is you can take them anytime during the time frame the link is available, and often you will be able to take them repeatedly with the highest score be recorded. The link (URL) for these will be posted on Canvas. **The lowest quiz scores will be drop when the quiz average is calculated.**

Assignments - Homework:
During the semester there will be multiple ~10 homework assignments. They will be announced in class and posted on Canvas. Homework assignments may be hand-written, if neat well organized, and written legibly. Pictures or scanning of the handwritten work must be import into a document (e.g. pdf/doc/etc) and must be neat an organized**.** Assignments will be graded for neatness, completeness, and effort. It is the students' responsibility to go over solutions to problems/exercises (if provided), ask questions and get doubts clarified on an on-going basis. All assignment must include at the top: **student name** (first and last-name), the **course name** (and section number), homework or project **assignment number**, and **date submitted**. The assignments will be posted with and link to submission on Canvas. All submission but be done on or before the date due and via a Canvas assignment link, never email assignments. **The lowest homework assignment will be dropped before the homework assignment average is calculated.**

Assignments – Coding Projects*:*
During the semester there will be multiple programming projects assigned. They will be announced and discussed in lecture and posted submission link posted on Canvas. These assignments may require the submissions to be uploaded to Canvas on or before the date due (never email assignments).

- The submissions for the projects must include source code.
- The source code must be submitted, as plain text and with the correct file ending (.java, .c, scm, .pl, …) never import into a PDF document (for example).
- **The submitted code must build (compile and link) under the specified language**.
  - **if you cannot get all feature to work, turn in your best working version (and clearly state what is not working)**
- The submitted code should give correct output.
  - submitting code that prints "hello world" for a project to find the fifth Finocchi number is obviously completely unacceptable (negative score).
- The submitted code must be clearly documented and well formatted. It is very important that your code be very easy to read. This last part is **very important**, your code will be read by the instructor, and readability count heavily in the grading.
- The submitted file must strictly adhere to the naming convention for the submitted files (**see below**).
- Submitting plagiarized code will be grading down harshly (see academic integrity). Plagiarism is representing someone else work as one's own, fraudulently trying to take credit for work one did not do. Plagiarized code can come from, for example, the Internet (cheg.com), classmate (providing working code is academic dishonesty) or Artificial Intelligence (ChatGPT) Always **type up** your own code. Working in groups and helping each other is good. But, do not, as a group, produce core of the solution that then everyone then uses (electronically copies) for their submission. These submissions may very likely be flagged as plagiarized code and graded as such. You must always type up your own solution in your own words**.** Electronically copying someone else's code but changing variable names is still Plagiarism. Allowing someone to copy your code is not in itself plagiarism but is academic dishonest. Looking at someone else's code (with permission), figuring out it works and then writing your own version is NOT Plagiarism. Showing someone how to code works is NOT Plagiarism.
- **The lowest project assignment score will be dropped before the project average is calculated.**

Assignments Submission Rules

- Always submit to the Canvas assignment link. NEVER submit via email (assignments send via email will be ignored). Make sure your assignment is exactly how you want it to be before submitting. For example, double check that it meets all specification before submitting.

- Any remarks, notes or comments should be put into the comment section of the Canvas assignment link (for example – what does not work)
- Only submit the file or files requested. Never submit IDE (e.g. Eclipse or NetBeans) project files. Submit only the source code files (no folder/directory)
- **Always** include the **file-ending** (.java) in the name. It is not sufficient to just type in correct name in the Canvas assignment (upload) link.

- For homework only submit files sof the following type
  - .txt, .doc, .dcx, pdf, … Never use **PAGES** (a mac word processor) or RTF (Rich Text format)
  - Unless asked do NOT submit pictures (see next item)
  - If you are aksed to submit pictures (or scanning) your handwritten homework, import them in a single document (e.g. pdf/doc/etc) and organize them (ie put them in order)
- Do not use archives (ZIP archive) unless explicitly asked to do so.
  - If asked to use zip, archive only the files requested (i.e. source code). Never include subdirectories.
  - Do not use rar (yes, rar is better then zip but use zip)
  - Never ever zip up a zip archive

- Source Code –
  - ALWAYS format and **indent** your code correctly.
    - correctly indented and spaced out so it is easy to read (warning I am a bit **OCD** about **indentations**)
  - ALWAYS add comments
    - a comment header at the top of each file with: name, course, project#, description,
    - comments for all methods procedures functions.

Grading and Scale:

Metrics:

Category	**total number**	**number used**	**total weigth**
Exam	**2**	**2**	**45%**
Projects	**n***	**n-1**	**35%**
Homework	**n***	**n-1**	**10%**
Quizzes	**n***	**n-1**	**10%**

Grading Scale:

A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
100-91	90	89	88-81	80	79	78-71	70	69	68-63	62	61-0

The grades for individual assignments, projects, quizzes, and exams will be **posted on Canvas**, but the grading facilities of Canvas' Grade Center will **not** be used. The column for 'Total Points' will not be correct, or even significantly correlated to your grade. Your grade is based on the above weighted categories

your_course_grade = (exam_ave * 0.45) + (project_ave * 0.35) + (hw_ave * 0.1) + (quizzes_ave * 0.1)

Extra Points: There may be extra credit offered during the semester. It will be added to the score for one of the categories (i.e. homework extra credit). Because it will then be possible to earn more the 100% of the possible point in one or more of the categories, there will be a cap of 100% for all categories

I will drop the lowest project and homework score and the lowest two quiz scores before calculating average (this is an example of something Canvas can not do).
Extra Points: There may be extra credit offered during the semester. It will be added to the score for one of the categories (i.e. homework extra credit). Because it will then be possible to earn more the 100% of the possible point in one or more of the categories, there will be a cap of 100% for all categories

I understand the terms of the grading policy and grading scale. _______________. initials/date.

Important Days
For more information on the campus' Academic Calendar, visit the **CSUDH Academic Calendar page.**

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Course Outline and Tentative Schedule

Week	Chapter/Topic
1	Review of CSC 121 topics (Chapters 1 – 7 and 15.1 – 2)
2	Chapter 8 A Second Look at Classes and Objects
3	Chapter 8 A Second Look at Classes and Objects
4	Chapter 9: Text Processing and More about Wrap- per Classes
5	Chapter 9 and Chapter 10: Inheritance
6	Chapter 10: Inheritance
7	Chapter 10: Inheritance
8	Review, and Exam
9	Chapter 11: Exceptions and Advanced File I/O (11.3 can be skipped)
10	Chapter 11: Exceptions and Advanced File I/O (11.3 can be skipped)
11	Chapter 15 Recursion
12	Chapter 15 Recursion
13	Chapter 12 JavaFX: GUI Programming and Basic Controls
14	Chapter 12 JavaFX: GUI Programming and Basic Controls
15	
16	**Final Exam**